

CODEBOOK for Reproduction of a Research Claim from Smith et al. (2016)

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Variables used in ERGM Estimation

We describe here the set of variables is used in the first-stage ERGM estimation. An additional set of variables used for the meta-analysis are described separately in below.

The primary variable of interest in the first stage relates to the estimation of a GENERAL HOMOPHILY parameter within classrooms. This is based on information on classroom friendship networks, and captures the tendency for adolescents to name same-ethnic peers as friends.

Ethnicity is coded on the basis of the national origin of students' parents (see Smith et al. pp.1238–1239). Native students are defined as adolescents whose biological parents were both born in the host country. Immigrant adolescents are likewise coded to their parents' country of origin. When adolescents have mixed heritage, they are coded to the immigrant parent in the case of a mixed native-immigrant background. In case parents originate from different nonnative countries, students are coded to the mother's country.

Several controls are also in the ERGMs (see p.1240):

1. A variable is constructed indicating whether dyads involve students of the same GENDER.
2. A second control relates to the POPULARITY DIFFERENCE between peers in a dyad. Note that students are asked to nominate that most popular students in the classroom. Following the code provided by the authors of the original study, we code the popularity score as the total number of incoming nominations divided by the size of the network. The POPULARITY DIFFERENCE is then calculated as the difference between popularity scores for the students in each dyad.
3. The SOCIOECONOMIC DIFFERENCE between peers in a dyad is calculated on the basis of the highest occupational status in an adolescent's household.¹ The higher of the mother / father ISEI scores is used.² If scores are missing for both parents, the mean ISEI score in the classroom is used. Once ISEI scores are assigned to all students in this manner, SOCIOECONOMIC DIFFERENCE is calculated as the absolute difference in scores between pairs in the dyad.

Beyond these survey-based variables, the ERGMs use several variables describing the network structure itself (see pp.1240–1241):

4. EDGES measures the total number of friendship ties and captures the baseline density of the network.
5. MUTUAL reflects the tendency of adolescents to reciprocate friendships and denotes instances where i and j both nominate each other as friends.
6. GWESP ("geometrically weighted edgewise shared partner") captures transitivity, or the fact that i and j are more likely to become friends when they have more friends in common.
7. GWESP θ captures decreasing marginal returns of additional shared friends. As explained by Smith et al. p.1241: GWESP θ reflects the intuition that "the more friends you share, the more likely a friendship is, but the more friends you share, the smaller the effect of every additional friend on friendship becomes."

¹As noted in our pre-registration document, Smith et al. constructed occupational status scores based on occupational categories, while we use the International Socio-Economic Index of Occupational Status (ISEI-08) Scores provided in the newest version of the CILS4EU dataset.

²Note also that CILS4EU actually contains both a student and a parental questionnaire, and ISEI scores for both mother and father can be obtained from both questionnaires. In case of discrepancies, we use information contained on the child questionnaire, if available. If this information is not available on the child questionnaire for either the mother or the father, we use the highest ISEI score as indicated in the parental questionnaire.

Variables used in the Meta-Analysis

Turning to the meta-analysis, the main explanatory variable in the model we are asked to replicate (Model 2, Table 3, Smith et al. p.1247) concerns the PROPORTION NATIVES in the classroom network. This variable also enters as a squared term in the model.

The meta-analysis also includes several additional controls:

8. CLASS SIZE
9. MEAN SOCIOECONOMIC STATUS, which is constructed by averaging the ISEI scores associated with all students in the classroom
10. Dummy variables for German, Dutch, and Swedish classes (with English classes as the reference category)

Citation: Smith, S., Van Tubergen, F., Maas, I., & McFarland, D. A. (2016). "Ethnic composition and friendship segregation: differential effects for adolescent natives and immigrants." *American Journal of Sociology*, 121(4), 1223-1272.